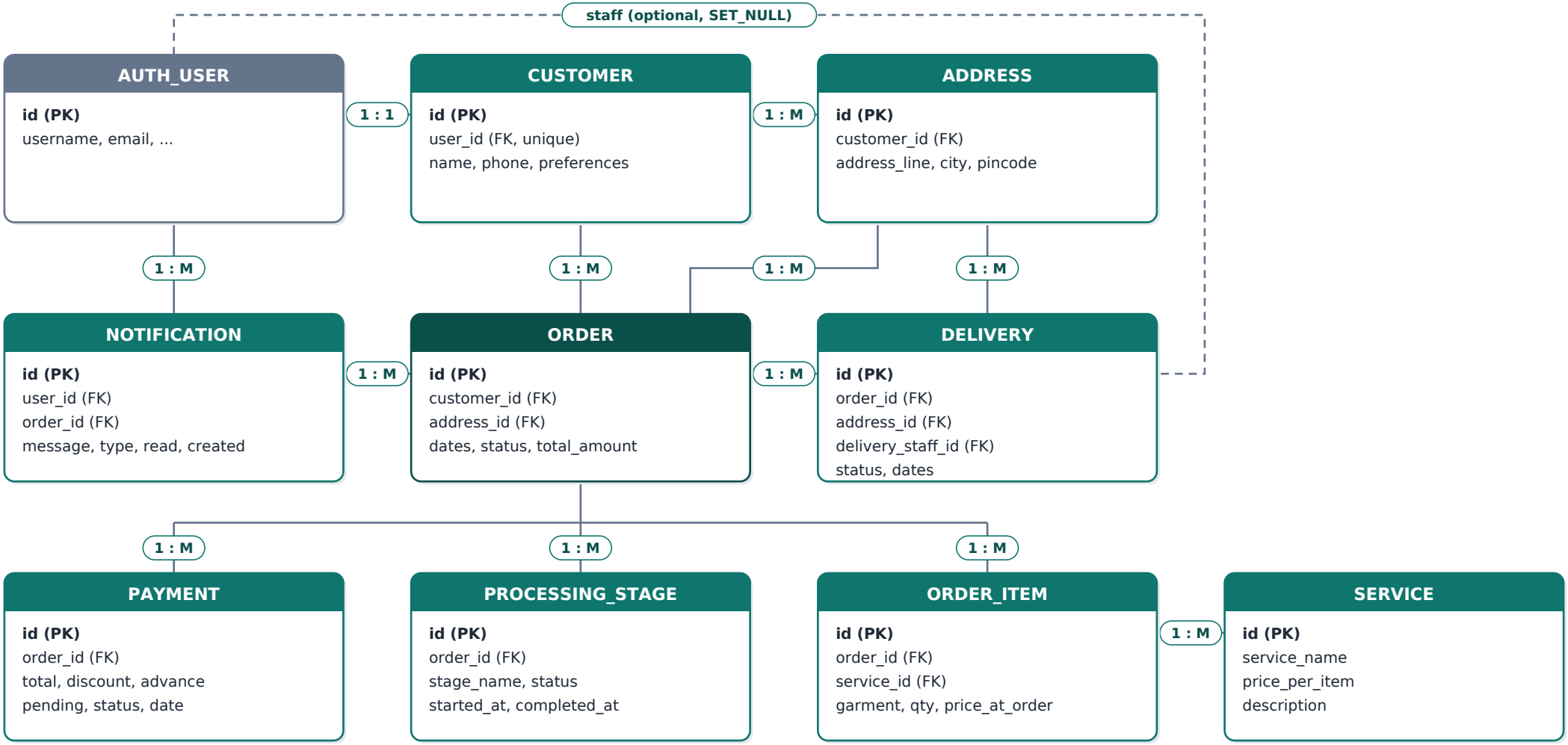
Entity Relationship Diagram (ERD)

Database structure documented from the supplied Django models.py. The built-in Django User model is shown as AUTH_USER.

LaundryCare

1. Entity relationship overview



Additional links: CUSTOMER to ORDER (1:M); ADDRESS to DELIVERY (1:M); AUTH_USER to DELIVERY.delivery_staff (optional, SET_NULL). AUTH_USER to NOTIFICATION (1:M). CASCADE applies to owned child records; PROTECT applies to referenced addresses and services.

2. Relationship and integrity notes

The diagram represents the project’s ten application models plus Django’s built-in authentication User model. FK labels describe the model relationships; the detailed table below lists the important cardinalities and deletion rules.

Relationship	Cardinality	Model rule / meaning
AUTH_USER → CUSTOMER	1 : 1	Customer.user is a OneToOneField with CASCADE.
CUSTOMER → ADDRESS	1 : many	Address.customer uses CASCADE; a customer can save multiple addresses.
CUSTOMER → ORDER	1 : many	Order.customer uses CASCADE; a customer can place multiple orders.
ADDRESS → ORDER	1 : many	Order.address uses PROTECT; an address referenced by an order cannot be deleted.
ORDER → ORDER_ITEM	1 : many	OrderItem.order uses CASCADE; each item belongs to one order.
SERVICE → ORDER_ITEM	1 : many	OrderItem.service uses PROTECT; the service is retained if referenced by an item.
ORDER → PROCESSING_STAGE	1 : many	ProcessingStage.order uses CASCADE; stages record order progress.
ORDER → DELIVERY	1 : many	Delivery.order uses CASCADE; delivery records are linked to an order.
ADDRESS → DELIVERY	1 : many	Delivery.address uses PROTECT.
AUTH_USER → DELIVERY	1 : many / optional	Delivery.delivery_staff uses SET_NULL and is nullable.
ORDER → PAYMENT	1 : many	Payment.order uses CASCADE; payment records store totals and payment status.
AUTH_USER → NOTIFICATION	1 : many	Notification.user uses CASCADE.
ORDER → NOTIFICATION	1 : many	Notification.order uses CASCADE.

Key fields by entity

Entity	Core fields (summary)
Customer	user (unique FK), name, phone, preferences
Address	customer FK, address_line, city, pincode
Service	service_name, price_per_item, description
Order	customer FK, address FK, order_date, expected_date, special_instructions, status, total_amount
OrderItem	order FK, service FK, garment_type, quantity, condition, price_at_order
ProcessingStage	order FK, stage_name, status, started_at, completed_at
Delivery	order FK, delivery_staff FK (nullable), pickup_date, delivery_date, delivery_status, address FK
Payment	order FK, total_amount, discount, advance_amount, pending_amount, payment_status, payment_date
Notification	user FK, order FK, message, notification_type, is_read, created_at
AUTH_USER	Django built-in authentication model; username and other standard auth fields